

Parth N. Kheni

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EDUCATION

Boston University

Boston, MA

B.S. Computer Engineering, Concentration in Machine Learning

Expected Dec 2026

- Related Coursework: Machine Learning, Probability Statistics & Data Science, Computation Linear Algebra, Applied Algorithms, Cloud Computing, Software Engineering, Computer Architecture, Differential Equations

Rutgers University

New Brunswick, NJ

B.S. Computer Engineering | GPA: 3.97 | Dean's List

Sept 2023 – May 2024

SKILLS

Programming: Python, SQL, MATLAB, C++ (primary), Java, C#, Objective-C, OOP

Libraries/Frameworks: Numpy, Matplotlib, Pandas, scikit-learn, TensorFlow, PyTorch

Cloud/Tools: Azure AD, GitHub, Microsoft 365, Arduino, Android Studio, API, Power BI, WorkManager

Systems & Analysis: Systems Integration, Cross-Team Coordination, Time-Series Analysis, Predictive Modeling

EXPERIENCE

Beth Israel Deaconess Medical Center (BIDMC), Machine Learning Intern

Sept 2025 – Present

- Lead a multi-institutional collaboration between BIDMC and Johns Hopkins across engineering and clinical teams to develop and integrate a learning-based autonomous wound closure algorithm within the da Vinci Research Kit (dVRK) platform for future clinical translation, coordinating requirements and experiment milestones.
- Build repeatable wound-closure experiments by standardizing task setup, ROS data capture, labeling guidelines, and evaluation metrics; maintain consistent dataset formats across runs to enable clean comparisons and faster iteration on autonomy behavior.
- Maintain a reproducible NVIDIA Isaac Sim + ORBIT-Surgical pipeline on BU's Shared Computing Cluster (SCC) using Singularity containers, and document protocols, results, and meeting action items to keep development aligned.

Boston University Center for Space Physics, Machine Learning Researcher – Space Weather ML

Apr 2025 – Present

- Build end-to-end Python pipelines for satellite instrument data (POES spacecraft) and solar wind measurements (OMNI), generating model-ready datasets with 7M+ 2-second samples per month for space weather forecasting.
- Train and evaluate PyTorch neural networks and Ridge regressors to forecast energetic electron precipitation (E4_0/E4_90), improving RMSE and correlation over prior baselines using standardized features and time-ordered train/val/test splits.
- Improve model generalization across storm conditions and radiation belt locations by evaluating temporal resolution trade-offs, quantile-based flux aggregations, and local-time features, with focus on rare high-flux event performance.

PUBLICATIONS & PRESENTATIONS

- Capannolo, L., Pettit, J., Elliott, S., Morales, J. P. B., Bibeau, A., **Kheni, P. N.** *Feb 2026*
“Monitoring radiation belt electrons at LEO: preliminary results of MAPLE (MAPs of LEO Electrons) model.”
COSPAR 2026 – Abstract (submitted)

PROJECTS

LLM-Assisted Incident Summarization & Clustering (Microsoft-affiliated), Applied ML Engineer

Jan 2026 – Present

- Build an end-to-end incident intelligence pipeline: ingest large-scale system/application logs, generate structured incident summaries with LLMs, embed summaries, and cluster incidents to identify duplicates and failure patterns.
- Implement a backend service and lightweight UI/API to explore incidents and clusters, making practical accuracy vs cost/latency tradeoffs to support scalable operation on noisy real-world logs.

Whack-a-Mole Reaction Game on FPGA (Nexys4 DDR, Verilog), FPGA Design Engineer

Sept 2025 – Dec 2025

- Built a real-time Whack-a-Mole game on a Nexys4 DDR FPGA with 30-second multi-difficulty rounds, 5 LED “moles”, and per-mole switch/hammer input, using Verilog for debounced I/O, timing, random selection.
- Improved reliability and integration, delivering a glitch-free demo by owning input and game-control modules, coordinating interfaces across a team, and iterating with Vivado simulation, timing analysis, and on-board testing.

Autonomous Line-Following Robot (MAPS), Embedded Systems Engineer

Jan 2025 – May 2025

- Built an autonomous robot using Arduino, L298N motor driver with PWM speed control, IR and ultrasonic sensor fusion, and real-time feedback loops, achieving >90% line-following accuracy with obstacle detections at 10 cm range.
- Designed custom 3D-printed PLA chassis, created circuit wiring schematic, and iterated through sensor calibration and control logic debugging across hardware and software.

AWARDS & HONORS

NSF ACCESS / Jetstream2 Allocation (CIS260066) – \$31,444 awarded; 146,000 GPU SUs

Jan 2026 – Present

GEM Workshop – Invited Speaker (Portland, ME) – selected to present machine learning research.

July 2026

BU Undergraduate Research Opportunities Program (UROP) Award – funded ML research.

Summer 2026

ACTIVITIES

BU Baja Society of Automotive Engineers, Head Engineer of Computer Systems

Sept 2024 – Present

BU Jalwa Dance Troupe, Secretary

Sept 2024 – Present